



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade, UG Programmes CE, CSE, ECE, EEE, IT & ME are Accredited by NBA
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

Regulation: R20			II / IV - B.Tech. I - Semester						
COMPUTER SCIENCE & ENGINEERING									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards)									
Course Code	Course Name	Category	Cr	L	T	P	Int. Marks	Ext. Marks	Total Marks
B20 BS 2101	Numerical Methods & Vector Calculus	BS	3	3	0	0	30	70	100
B20 CS 2101	Object Oriented Programming through C++	PC	3	3	0	0	30	70	100
B20 CS 2102	Software Engineering	PC	3	3	0	0	30	70	100
B20 CS 2103	Operating Systems	PC	3	3	0	0	30	70	100
B20 CS 2104	Python Programming	PC	3	3	0	0	30	70	100
B20 CS 2105	Object Oriented Programming Lab	PC	1.5	0	0	3	15	35	50
B20 CS 2106	Python Programming Lab	PC	1.5	0	0	3	15	35	50
B20 CS 2107	Operating Systems Lab	PC	1.5	0	0	3	15	35	50
B20 CS 2108	Web Application Development Using Full Stack	SOC	2	0	0	4	--	50	50
B20 MC 2101	Environmental Science	MC	0	2	0	0	--	--	--
B20 MC 2103	English Proficiency	MC	0	2	0	0	--	--	--
TOTAL			21.5	19	0	13	195	505	700

Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS2101	BS	3	--	--	3	30	70	3 Hrs.
NUMERICAL METHODS & VECTOR CALCULUS								
COMMON TO CE, CSE, EEE&IT								
Pre-requisites: Basic concepts of calculus.								
Course Objectives: Students are expected to learn								
1	Numerical methods to solve algebraic and transcendental equations and the concept of interpolation and its use for equally and unequally spaced data points,							
2	Methods for numerical evaluation of integrals and for solving first order ODEs.							
3	Concepts of double, triple integrals and its applications.							
4	Concepts of Gradient, divergence, curl.							
5	Vector integral theorems.							
Course Out Comes: At the end of the course students will be able to								
S. No	OUT COME							Knowledge Level
1	Determine a real root of an algebraic or transcendental equation. Fit an interpolation formula and perform interpolation for equally spaced and unequally spaced data.							K3
2	Evaluate numerically certain definite integrals. Solve a first order ordinary differential equation by Euler and RK methods.							K3
3	Evaluate double integrals and determine the areas.							K3
4	Evaluate triple integrals and determine the volumes.							K3
5	Find the gradient of a scalar function, divergence and curl of a vector function.							K3
6	Solve simple problems using vector integral theorems.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Solution of Algebraic and Transcendental Equations: Introduction, Bisection method, Method of false position, Iteration method &Newton-Raphson method. Interpolation: Introduction, forward differences, backward differences, and Central differences. Differences of a polynomial, Newton’s forward, and backward interpolation formulae. Interpolation with unequal intervals:Newton’s divided difference and Lagrange interpolation formulae.							
UNIT-II (10 Hrs)	Numerical Integration and solution of Ordinary Differential equations: Trapezoidal rule, Simpson’s 1/3 rd rule. Solution of first order ordinary differential equations subjected to initial conditions by Taylor’s method, Picard’s method, Euler’s method, Modified Euler’s method and Fourth order Runge-Kutta method.							

UNIT-III (12 Hrs)	Multiple integrals: Double and triple integrals, Change of order of integration. Change of variables, applications to finding Areas and Volumes.
UNIT-IV	Vector differentiation: Scalar and vector point functions, Vector Differentiation, Gradient, Directional derivative, Divergence, Curl, Scalar Potential.
UNIT-V (14Hrs)	Vector Integration: Line integral, Work done; Area, Surface and volume integrals, Vector integral theorems: Greens, Stokes, and Gauss Divergence theorems (without proof).
TEXTBOOKS:	
1.	Scope and Treatment as in “Higher Engineering Mathematics”, by Dr.B.S.Grewal, 43 rd Edition, Khanna Publishers.
REFERENCE BOOKS:	
1.	Advanced Engineering Mathematics, by Erwin Kreyszig, Wiley
2.	Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
3.	A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
4.	Peter O’ Neil, Advanced Engineering Mathematics, Cengage.
5.	Advanced Engineering Mathematics, by H.K.Dass, S.ChandCompany.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS2101	PC	3	0	0	3	30	70	3 Hrs.
OBJECT ORIENTED PROGRAMMING THROUGH C++								
(For CSE)								
Course Objectives:								
1.	Understand the syntax and principles of Object Oriented Programming.							
2.	Design and development of secure and extendable C++ applications.							
3.	Describe the concept of function overloading, operator overloading, virtual functions and polymorphism.							
4.	Classify inheritance with the understanding of early and late binding, usage of exception handling, generic programming.							
5.	Demonstrate the use of various OOP's concepts with the help of programs.							
Course Outcomes: At the end of the course Students will be able to								
1.	Illustrate the process of Object Oriented Paradigm.							K2
2.	Demonstrate classes, member functions, constructors and their importance in developing real world applications.							K2
3.	Apply C++ features such as Inheritance, operator overloading to make programs reusable.							K3
4.	Understand Dynamic Memory Management techniques using pointers.							K2
5.	Apply the concept of Generic Programming and Exception handling to build an efficient and error free code.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to C++: Difference between C and C++, Evolution of C++, The Object Oriented Technology, Disadvantage of Conventional Programming, Key Concepts of Object Oriented Programming, Advantages of OOP's, Object Oriented Language.							
UNIT-II (10 Hrs)	Classes and Objects: Classes in C++, Declaring Objects, Access Specifiers and their Scope, Defining Member Function, Overloading Member Function, Nested class. Constructors and Destructors: Introduction, Constructors and Destructor, Characteristics of Constructor and Destructor, Application with Constructor, Constructor with Arguments, Parameterized Constructor, Destructors, Anonymous Objects							
UNIT-III (10 Hrs)	Operator Overloading, Type Conversion and Inheritance: The Keyword Operator, Overloading Unary Operator, Operator Return Type, Overloading Assignment Operator (=), Rules for Overloading Operators, Inheritance, Reusability, Types of Inheritance, Virtual Base Classes, Object as a Class Member, Abstract Classes, Advantages of Inheritance, Disadvantages of Inheritance.							
UNIT-IV (10 Hrs)	Pointers: Pointer, Features of Pointers, Pointer Declaration, Pointer to Class, Pointer Object, The this Pointer, Pointer to Derived Classes and Base Class. Binding Polymorphisms and Virtual Functions: Introduction, Binding in C++, Virtual Functions, Rules for Virtual Function, Virtual Destructor.							

UNIT-V (10 Hrs)	Generic Programming with Templates & Exception Handling: Definition of class Templates, Normal Function Templates, Over Loading of Template Function, Bubble Sort Using Function Templates, Difference between Templates and Macros, Linked Lists with Templates, Exception Handling, Principles of Exception Handling, and The Keywords- try throw and catch, Multiple Catch Statements, Specifying Exceptions. Overview of Standard Template Library, STL Programming Model, Containers, Sequence Containers, Associative Containers, Algorithms, Iterators, Vectors, Lists, Maps.
Text Books:	
1.	A First Book of C++, 4 th Edition, Gary Bronson, Cengage Learning.
2.	The Complete Reference, C++, 5 th Edition, Herbert Schildt, McGraw-Hill Education.
Reference Books:	
1.	Object Oriented Programming C++, Joyce Farrell, Cengage Learning.
2.	C++ Programming: from problem analysis to program design, 6th Edition, DS Malik, Cengage Learning
3.	Programming in C++, Ashok N Kamthane, and Pearson.
4.	Object Oriented Programming using C++, 8th Edition, E.Balaguruswamy, PHI
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105151/
2.	https://github.com/topics/object-oriented-programming

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS2102	PC	3	0	0	3	30	70	3 Hrs.
SOFTWARE ENGINEERING								
(For CSE)								
Course Objectives:								
1.	Give exposure to phases of Software Development, common process models including Waterfall, the Unified Process, and elements of the Agile Process.							
2.	Give exposure to a variety of Software Engineering practices such as Requirements Analysis and Specification.							
3.	Give exposure to Software Design Techniques.							
4.	Give exposure to various Software Quality Assurance and Testing strategies.							
Course Outcomes:								
At the end of the course Students will be able								
1.	Understand different software process models and their significance.							K2
2.	Distinguish various requirements identification procedures.							K2
3.	Identify different methods for requirement analysis modeling.							K2
4.	Differentiate various aspects of system design and software architectures.							K2
5.	Apply software quality assurance and testing strategies.							K3
SYLLABUS								
UNIT-I (11Hrs)	The Nature of Software, The Unique Nature of WebApps, Software Engineering, The Software Process, Software Engineering Practice, Software Myths. A Generic Process Model, Process Assessment and Improvement, Prescriptive Process Models, Specialized Process Models, The Unified Process, Agility, Agility and the Cost of Change, Agile Process, Extreme Programming (XP), Other Agile Process Models.							
UNIT-II (8Hrs)	Requirements Engineering, Establishing the Groundwork, Eliciting Requirements, Developing Use Cases, Building the Requirements Model, Negotiating Requirements, Validating Requirements, Requirements Analysis.							
UNIT-III (09Hrs)	Scenario-Based Modeling, UML Models ThatSupplement the Use Case, Data Modeling Concepts, Class-Based Modeling,Requirements Modeling Strategies, Flow-Oriented Modeling, Creating a Behavioral Model, Patterns for Requirements Modeling, Requirements Modeling forWebApps.							
UNIT-IV (13Hrs)	Design within the Context of Software Engineering, The Design Process, Design Concepts, The Design Model, Software Architecture, Architectural Genres, Architectural Styles, Assessing Alternative Architectural Designs, Architectural Mapping Using Data Flow, Components, Designing Class-Based Components, Conducting Component-Level Design, Component-Level Design for Web Apps, Designing Traditional Components, Component-Based Development.							

	The Golden Rules, User Interface Analysis and Design, Interface Analysis, Interface Design Steps, WebAppInterface Design, Design Evaluation.
UNIT-V (09Hrs)	Elements of Software Quality Assurance, SQA Tasks, Goals & Metrics, Statistical SQA, Software Reliability, A Strategic Approach to Software Testing, Strategic Issues, Test StrategiesforConventionalSoftware,TestStrategiesforObject-OrientedSoftware, Test Strategies for WebApps, Validation Testing, System Testing, The Art of Debugging, Software Testing Fundamentals, Internal and External Viewsof Testing, White-Box Testing, Basis Path Testing.
Text Books:	
1.	Software Engineering: A Practitioner's approach, Roger S Pressman, 7th edition McGraw Hill Higher Education (2009)
2.	Software Engineering, Ian Sommerville, 9th edition. Pearson (2017)
Reference Books:	
1.	Software Engineering, A Precise Approach, PankajJalote, Wiley India, 2010.
2.	Software Engineering, UgrasenSuman, Cengage (2012)
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105182/

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS2103	PC	3	0	0	3	30	70	3 Hrs.
OPERATING SYSTEMS								
(For CSE)								
Course Objectives:								
1.	Introduce to the internal operation of modern operating systems							
2.	Define, explain, processes and threads, mutual exclusion, CPU scheduling, deadlock, memory management, and file systems							
3.	Understand File Systems in Operating System like UNIX/Linux and Windows							
4.	Understand Input Output Management and use of Device Driver and Secondary Storage (Disk) Mechanism							
5	Analyze Security and Protection Mechanism in Operating System							
Course Outcomes:								
At the end of the course Students will be able								
1.	Describe various generations of Operating System and functions of Operating System, System calls							K2
2.	Describe the concept of process, threads and analyze various CPU Scheduling Algorithms and IPC							K2
3.	Illustrate memory management strategies							K3
4.	illustrate deadlocks, files and Secondary-Storage Structure							K3
5.	Summarize Security and Protection Mechanism in Operating Systems. Understand the Operating System like UNIX/Linux and Windows							K3
SYLLABUS								
UNIT-I (6Hrs)	Operating Systems Overview: Operating system functions, Operating system structure, Operating systems operations, Computing environments, Open-Source Operating Systems. System Structures: Operating System Services, User and Operating-System Interface, systems calls, Types of System Calls, system programs, operating system structure, operating system debugging, System Boot.							
UNIT-II (10Hrs)	Process Concept: Process scheduling, Operations on processes, Inter-process communication, Communication in client server systems. Multithreaded Programming: Multithreading models, Thread libraries, Threading issues. Process Scheduling: Basic concepts, Scheduling criteria, Scheduling algorithms, Multiple processor scheduling, Thread scheduling. Inter-process Communication: Race conditions, Critical Regions, Mutual exclusion with busy waiting, Sleep and wakeup, Semaphores, Mutexes, Monitors, Message passing, Barriers, Classical IPC Problems - Dining philosophers problem, Readers and writers problem.							

UNIT-III (06Hrs)	Memory-Management Strategies: Introduction, Swapping, Contiguous memory allocation, Paging, Segmentation. Virtual Memory Management: Introduction, Demand paging, Copy on-write, Page replacement, Page replacement Algorithms, Frame allocation, Thrashing, Memory-mapped files, Kernel memory allocation.
UNIT-IV (10Hrs)	Deadlocks: Resources, Conditions for resource deadlocks, Deadlock detection and recovery, Deadlock avoidance, Deadlock prevention. File Systems: Files, Directories, File system implementation, management and optimization. Secondary-Storage Structure: Overview of disk structure, and attachment, Disk scheduling, RAID structure, Stable storage implementation.
UNIT-V (08Hrs)	System Protection: Goals of protection, Principles and domain of protection, Access matrix, Access control, Revocation of access rights. System Security: Introduction, Program threats, System and network threats. Case Studies: Linux, Microsoft Windows.
Text Books:	
1.	Silberschatz A, Galvin P B, and Gagne G, Operating System Concepts, 9th edition, Wiley, 2013.
2.	Tanenbaum A S, Modern Operating Systems, 3rd edition, Pearson Education, 2008. (for Interprocess Communication and File systems.)
Reference Books:	
1.	Dhamdhere D M, Operating Systems A Concept Based Approach, 3rd edition, Tata McGraw-Hill, 2012.
2.	Stallings W, Operating Systems -Internals and Design Principles, 6th edition, Pearson Education, 2009
3.	Nutt G, Operating Systems, 3rd edition, Pearson Education, 2004
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105214/

Code	Category	L	T	P	C	I.M	E.M	Exam
B20 CS 2104	PC	3	0	0	3	30	70	3 Hrs.
PYTHON PROGRAMMING								
(For CSE)								
Course Objectives:								
1.	To learn about Python syntax, semantics, and the runtime environment.							
2.	To learn the use of lists, tuples, dictionaries and sets in Python programs.							
3.	To learn the python package building and Python modules for reusability.							
4.	To be familiarized in general coding techniques and object-oriented programming.							
5	To develop the skills of designing GUI and handling exceptions in python.							
Course Outcomes:								
At the end of the course Students will be able								
1.	Understand the basic principles of python programming.							K2
2.	Apply the knowledge of python programming to perform operations on data structures.							K3
3.	Solve the coding tasks using functions and modular programming.							K3
4.	Use OOP principles and File concepts to solve different problems.							K3
5.	Handle different exceptions raised in python and apply GUI for providing interface to various problems.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction: Introduction to Python, Program Development Cycle, Input, Processing, and Output, Displaying Output with the Print Function, Comments, Variables, Reading Input from the Keyboard, Performing Calculations, Operators. Type conversions, Expressions, More about Data Output. Data Types, and Expression: Strings Assignment, and Comment, Numeric Data Types and Character Sets, Using functions and Modules. Decision Structures and Boolean Logic: if, if-else, if-elif-else Statements, Nested Decision Structures, Comparing Strings, Logical Operators, Boolean Variables. Repetition Structures: Introduction, while loop, for loop, Input Validation Loops, Nested Loops.							
UNIT-II (10Hrs)	Strings and Text Files: Accessing Character and Substring in Strings, Strings and Number Systems, String Methods, Text Files. Data structures: Lists- creating a list, accessing, slicing and other operations Tuples- creating a tuple, accessing and other operations Dictionaries- creating a dictionary, accessing keys and values and other operations Sets- creating a set, modifying, removing and other operations							
UNIT-III (10Hrs)	Design with Function: Functions as Abstraction Mechanisms, Problem Solving with Top Down Design, Design with Recursive Functions, Case Study Gathering Information from a File System, Managing a Program’s Namespace, Higher Order Function. Modules: Modules, Standard Modules, Packages.							

UNIT-IV (10Hrs)	File Operations: Reading config files in python, Writing log files in python, Understanding read functions, read(), readline() and readlines(), Understanding write functions, write() and writelines(), Manipulating file pointer using seek, Programming using file operations Object Oriented Programming: Concept of class, object and instances, Constructor, class attributes and destructors, Inheritance , overlapping and overloading operators, Adding and retrieving dynamic attributes of classes
UNIT-V (10Hrs)	Errors and Exceptions: Syntax Errors, Exceptions, Handling Exceptions, Raising Exceptions, User-defined Exceptions, Defining Clean-up Actions Graphical User Interfaces: The Behaviour of Terminal Based Programs and GUI -Based, Programs, Coding Simple GUI-Based Programs, Other Useful GUI Resources.
Text Books:	
1.	Fundamentals of Python First Programs, Kenneth. A. Lambert, 2 nd Edition, Cenagage learning, 2018.
2.	Python Programming: A Modern Approach, VamsiKurama, Pearson,2018.
Reference Books:	
1.	Introduction to Python Programming, Gowrishankar.S, Veena A, first edition ,CRC Press,2018.
2.	Introduction to Programming Using Python, Y. Daniel Liang, Pearson,2013.
e-Resources:	
1.	https://www.tutorialspoint.com/python3/python_tutorial.pdf

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS2105	PC	0	0	3	1.5	15	35	3 Hrs.
OBJECT ORIENTED PROGRAMMING LAB								
(For CSE)								
Course Objectives: The objective of this lab is to								
1.	Demonstrate procedural and object oriented paradigm with concepts of streams, classes, functions, data and objects.							
2.	Understand dynamic memory management techniques using pointers, constructors, destructors, etc.							
3.	Demonstrate the concept of function overloading, operator overloading, virtual functions and polymorphism, inheritance.							
Course Outcomes: At the end of the course Students will be able								
1.	Able to develop simple programs using classes, objects and constructors in C++.							K3
2.	Able to implement Inheritance and polymorphism in C++							K3
3.	Implement Object Oriented Programs using Templates and Exception Handling.							K3
LIST OF PROGRAMS								
1	Exercise - 1 (Classes Objects) 1. Write a program to implement stack operations using classes and objects. 2. Write a C++ Program to illustrate various Constructors. 3. Write a C++ Program to illustrate Destructor. 4. Write a program for illustrating function overloading.							
2	Exercise – 2 (Access) 1. Write a program implementing Friend Function. 2. Write a program to illustrate <i>this</i> pointer. 3. Write a Program to illustrate pointer to a class.							
3	Exercise - 3 (Operator Overloading) 1. Write a program to Overload Unary Operator using Member Function. 2. Write a program to Overload Unary Operator using Friend Function. 3. Write a program to Overload Binary Operator using Member Function. 4. Write a program to Overload Binary Operator using Friend Function 5. Write a C ++ program to overload assignment operator (=)							
4	Exercise - 4 (Inheritance) 1. Write C++ Programs and incorporating various forms of Inheritance i) Single Inheritance ii) Hierarchical Inheritance iii)Multiple Inheritances iv)Multi-level inheritance v) Hybrid inheritance 2. Also illustrate the order of execution of constructors and destructors in inheritance							
5	Exercise - 5(Templates, Exception Handling) 1. Write a C++ Program to illustrate class Templates. 2. Write a C++ Program to illustrate Function Templates.							

	3. Write a C++ Program to illustrate Member Function Templates. 4. Write a Program for Exception Handling.
6	Exercise - 6 (Polymorphism) 1. Write a C++ program illustrating user defined string processing functions using pointers (string length, string copy, string concatenation) 2. Write a C++ program illustrating Virtual Base Classes. 3. Write a C++ Program illustrating Virtual functions.
Reference Books:	
1.	A First Book of C++, 4 th Edition, Gary Bronson, Cengage Learning.
2.	The Complete Reference, C++, 5 th Edition, Herbert Schildt, McGraw-Hill Education.
3.	Object Oriented Programming C++, Joyce Farrell, Cengage Learning.
4.	C++ Programming: from problem analysis to program design, 6th Edition, DS Malik, Cengage Learning.
5.	Programming in C++, Ashok N Kamthane, and Pearson.
6.	Object Oriented Programming using C++, 8th Edition, E.Balaguruswamy, PHI
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105151/
2.	https://github.com/topics/object-oriented-programming

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS2106	PC	0	0	3	1.5	15	35	3 Hrs.
PYTHON PROGRAMMING LAB								
(For CSE)								
Course Objectives: The Objectives of Python Programming are								
1.	To learn about Python programming language syntax, semantics, and the runtime environment							
2.	To be familiarized with universal computer programming concepts like data types, containers							
3.	To be familiarized with general computer programming concepts like conditional execution, loops & functions							
4.	To be familiarized with general coding techniques and object-oriented programming							
Course Outcomes:								
At the end of the course Students will be able								
1.	Apply the basics of programming in the Python language							K3
2.	Solve coding tasks related to conditional execution, and loops							K3
3.	Implement the operations of different data structures.							K3
4.	Solve coding tasks related to the fundamental notions and techniques used in object-oriented programming							K3
LIST OF PROGRAMS								
1	Write a program that asks the user for a weight in kilograms and converts it to pounds. There are 2.2 pounds in a kilogram.							
2	Write a program that asks the user to enter three numbers (use three separate input statements). Create variables called total and average that hold the sum and average of the three numbers and print out the values of total and average.							
3	Write a program that uses a <i>for</i> loop to print the numbers 8, 11, 14, 17, 20. . . 83, 86, 89.							
4	Write a program that asks the user for their name and how many times to print it. The program should print out the user’s name the specified number of times.							
5	Use a for loop to print a triangle like the one below. Allow the user to specify how high the triangle should be. * ** *** ****							
6	Generate a random number between 1 and 10. Ask the user to guess the number and print a message based on whether they get it right or not.							
7.	Write a program that asks the user for two numbers and prints <i>Close</i> if the numbers are within .001 of each other and Not close otherwise.							
8.	Write a program that asks the user to enter a word and prints out whether that word contains any vowels.							
9.	Write a program that asks the user to enter two strings of the same length. The program should then check to see if the strings are of the same length. If they are not, the program should print an appropriate message and exit. If they are of the same length, the program should alternate the							

	characters of the two strings. For example, if the user enters <i>abcde</i> and <i>ABCDE</i> the program should print out <i>AaBbCcDdEe</i> .
10.	Write a program that asks the user for a large integer and inserts commas into it according to the standard American convention for commas in large numbers. For instance, if the user enters 1000000, the output should be 1,000,000.
11.	In algebraic expressions, the symbol for multiplication is often left out, as in $3x+4y$ or $3(x+5)$. Computers prefer those expressions to include the multiplication symbol, like $3*x+4*y$ or $3*(x+5)$. Write a program that asks the user for an algebraic expression and then inserts multiplication symbols where appropriate.
12.	Write a program that generates a list of 20 random numbers between 1 and 100. (a) Print the list. (b) Print the average of the elements in the list. (c) Print the largest and smallest values in the list. (d) Print the second largest and second smallest entries in the list (e) Print how many even numbers are in the list.
13.	Write a program that asks the user for an integer and creates a list that consists of the factors of that integer.
14.	Write a program that generates 100 random integers that are either 0 or 1. Then find the longest run of zeros, the largest number of zeros in a row. For instance, the longest run of zeros in [1,0,1,1,0,0,0,0,1,0,0] is 4.
15.	Write a program that removes any repeated items from a list so that each item appears at most once. For instance, the list [1,1,2,3,4,3,0,0] would become [1,2,3,4,0].
16.	Write a program that asks the user to enter a length in feet. The program should then give the user the option to convert from feet into inches, yards, miles, millimeters, centimeters, meters, or kilometers. Say if the user enters a 1, then the program converts to inches, if they enter a 2, then the program converts to yards, etc. While this can be done with if statements, it is much shorter with lists and it is also easier to add new conversions if you use lists.
17.	Write a function called <i>sum_digits</i> that is given an integer num and returns the sum of the digits of num.
18.	Write a function called <i>first_diff</i> that is given two strings and returns the first location in which the strings differ. If the strings are identical, it should return -1.
19.	Write a function called <i>number_of_factors</i> that takes an integer and returns how many factors the number has.
20.	Write a function called <i>is_sorted</i> that is given a list and returns True if the list is sorted and False otherwise.
21.	Write a function called <i>root</i> that is given a number x and an integer n and returns $x^{1/n}$. In the function definition, set the default value of n to 2.
22.	Write a function called <i>primes</i> that is given a number n and returns a list of the first n primes. Let the default value of n be 100.
23.	Write a function called <i>merge</i> that takes two already sorted lists of possibly different lengths, and merges them into a single sorted list. (a) Do this using the sort method. (b) Do this without using the sort method.
24.	Write a program that asks the user for a word and finds all the smaller words that can be made from the letters of that word. The number of occurrences of a letter in a smaller word can't exceed the number of occurrences of the letter in the user's word.

25.	Write a program that reads a file consisting of email addresses, each on its own line. Your program should print out a string consisting of those email addresses separated by semicolons.
26.	Write a program that reads a list of temperatures from a file called <i>temps.txt</i> , converts those temperatures to Fahrenheit, and writes the results to a file called <i>ftemps.txt</i> .
27.	Write a class called Product. The class should have fields called name, amount, and price, holding the product's name, the number of items of that product in stock, and the regular price of the product. There should be a method <code>get_price</code> that receives the number of items to be bought and returns a the cost of buying that many items, where the regular price is charged for orders of less than 10 items, a 10% discount is applied for orders of between 10 and 99 items, and a 20% discount is applied for orders of 100 or more items. There should also be a method called <code>make_purchase</code> that receives the number of items to be bought and decreases amount by that much.
28.	Write a class called Time whose only field is a time in seconds. It should have a method called <code>convert_to_minutes</code> that returns a string of minutes and seconds formatted as in the following example: if seconds is 230, the method should return '5:50'. It should also have a method called <code>convert_to_hours</code> that returns a string of hours, minutes, and seconds formatted analogously to the previous method.
29.	Write a class called Converter. The user will pass a length and a unit when declaring an object from the class—for example, <code>c = Converter(9,'inches')</code> . The possible units are inches, feet, yards, miles, kilometers, meters, centimeters, and millimeters. For each of these units there should be a method that returns the length converted into those units. For example, using the Converter object created above, the user could call <code>c.feet()</code> and should get 0.75 as the result.
30.	Write a Python class to implement <code>pow(x, n)</code> .
31.	Write a Python class to reverse a string word by word.
32.	Write a program that opens a file dialog that allows you to select a text file. The program then displays the contents of the file in a textbox.
33.	Write a program to demonstrate Try/except/else.
34.	Write a program to demonstrate try/finally and with/as.
Reference Books:	
1.	Core Python Programming, Wesley J Chen, 2 nd edition, Pearson, 2007
2.	Python Data Science, Handbook, Essential tools for working with data, Jake VanderPlas,1 edition, Oreilly Publisher ,2016.
3.	Programming in Python 3--A Complete Introduction to the Python Language, Mark Summerfield, Second Edition, Addison Wesley, 2009.
4.	Getting Started with Python Data Analysis, Phuong Vo.T.H, Martin Czygan, Packt Publishing Ltd ,2015.
5.	Python Data Analysis, Ivan Idris,,3 rd edition, Packt Publishing Ltd .2021.
6.	Mastering Python Data Analysis, Magnus VilhelmPersson and Luiz Felipe Martins, Packt Publishing Ltd,2016.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20 CS 2107	PC	0	0	3	1.5	15	35	3 Hrs.
OPERATING SYSTEMS LAB								
(For CSE)								
Course Objectives:The objective of this lab is to								
1.	To understand the design aspects of operating system							
2.	To study the process management concepts & Techniques							
3.	To study the storage management concepts							
4.	To familiarize students with the Linux environment							
5.	To learn the fundamentals of shell scripting/programming							
Course Outcomes:								
At the end of the course Students will be able								
1.	To use Unix utilities and perform basic shell control of the utilities							K1
2.	To use the Unix file system and file access control							K2
3.	To use of an operating system to develop software							K2
4.	Students will be able to use Linux environment efficiently							K2
5.	Solve problems using bash for shell scripting							K2
LIST OF EXPERIMENTS								
1	a) Study of Unix/Linux general purpose utility command list: man,who,cat, cd, cp, ps, ls, mv, rm, mkdir, rmdir, echo, more, date, time, kill, history, chmod, chown, finger, pwd, cal, logout, shutdown. b) Study of vi editor c) Study of Bash shell, Bourne shell and C shell in Unix/Linux operating system d) Study of Unix/Linux file system (tree structure) e) Study of .bashrc, /etc/bashrc and Environment variables.							
2	Write a C program that makes a copy of a file using standard I/O, and system calls.							
3	Write a C program to emulate the UNIX ls -l command.							
4	Write a C program that illustrates how to execute two commands concurrently with a command pipe. Ex: - ls -l sort Also illustrate the order of execution of constructors and destructors in inheritance							
5	Simulate the following CPU scheduling algorithms: (a) Round Robin (b) SJF (c) FCFS (d) Priority Write a C++ Program to illustrate Member Function Templates.							
6	Multiprogramming-Memory management-Implementation of fork (), wait (), exec() and exit (), System calls							
7	Simulate the following: a) Multiprogramming with a fixed number of tasks (MFT) b) Multiprogramming with a variable number of tasks (MVT)							
8	Simulate Bankers Algorithm for Dead Lock Avoidance							

9	Simulate Bankers Algorithm for Dead Lock Prevention.
10	Simulate the following page replacement algorithms: a) FIFO b) LRU c) LFU
11	Simulate the following File allocation strategies (a) Sequenced (b) Indexed (c) Linked
12	Write a C program that illustrates two processes communicating using shared memory
13	Write a C program to simulate producer and consumer problem using semaphores
14	Write C program to create a thread using pthreads library and let it run its function.
15	Write a C program to illustrate concurrent execution of threads using pthreads library
Reference Books:	
1.	Unix concepts and applications by Sumitabha Das, TMH Publications.
2.	Unix programming by Stevens, Pearson Education.
3.	Shell programming by Yashwanth Kanetkar..
4.	Operating System Concepts by Silberschatz, and Peter Galvin

Code	Category	L	T	P	C	I.M	E.M	Exam
B20SO2201	SOC	0	0	4	2	--	50	3 Hrs.
WEB APPLICATION DEVELOPMENT USING FULL STACK								
(For CSE)								
FRONT END DEVELOPMENT MODULE-1								
Course Objectives: The objective of this lab is to								
1.	The objective of this lab is to provide understanding about the core concepts of frontend programming for web application.							
Course Outcomes:								
At the end of the course Students will be able								
1.	Analyze a web page and identify its elements and attributes.							K3
2.	Demonstrate the important HTML tags for designing static pages and separate design from content using Cascading Style sheet							K2
3.	Implement MVC and responsive design to scale well across PC, tablet and Mobile Phone.							K3
4.	Create web pages using HTML and Cascading Style Sheets.							K3
LIST OF EXPERIMENTS								
1	HTML 1) Introduction to HTML 2) Browsers and HTML 3) Editor’s Offline and Online 4) Tags, Attribute and Elements 5) Doctype Element 6) Comments 7) Headings, Paragraphs, and Formatting Text 8) Lists and Links 9) Images and Tables							
2	CSS 1) Introduction CSS 2) Applying CSS to HTML 3) Selectors, Properties and Values 4) CSS Colors and Backgrounds 5) CSS Box Model 6) CSS Margins, Padding, and Borders 7) CSS Text and Font Properties 8) CSS General Topics							

FRONT END DEVELOPMENT MODULE - 2		
Course Objectives:		
1.	The objective of this lab is to build strong foundation of JavaScript which will help developerto apply JavaScript concepts for responsive web frontend development	
Course Outcomes:		
At the end of the course Students will be able		
1.	Develop of the major Web application tier- Client side development	K3
2.	Participate in the active development of cross-browser applications through JavaScript	K3
3.	Develop JavaScript applications that transition between states	K3
LIST OF PROGRAMS		
Perform experiments related to the following concepts:		
1.	Introduction to JavaScript	
2.	Applying JavaScript (internal and external)	
3.	Understanding JS Syntax	
4.	Introduction to Document and Window Object	
5.	Variables and Operators	
6.	Data Types and Num Type Conversion	
7.	Math and String Manipulation	
8.	Objects and Arrays	
9.	Date and Time	
10.	Conditional Statements	
11.	Switch Case	
12.	Looping in JS	
13.	Functions	
Reference Books:		
1.	Programming the World Wide Web, 7th Edition Robet W Sebesta, Pearson, 2013.	
2.	WebTechnologies,1st Edition 7th impression, Uttam K Roy, Oxford, 2012.	
3.	Mean Stack Development,1st Edition, ELadElrom, A press O'Reilly,2016	
4.	Script & jQuery the missing manual, 2nd Edition, David sawyer mcfarland, O'Reilly, 2011.	
5.	Java 5. Web Hosting for Dummies, 1st Edition, Peter Pollock, John Wiley & Sons, 2013.	
6.	RESTful web services, 1st Edition, Leonard Richardson, Ruby, O'Reilly, 2007.	

Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC2101	MC	3	--	--	--	--	--	--
ENVIRONMENTAL SCIENCE								
(Common to CSE, ECE & IT)								
Course Objectives: The objectives of the course are to impart:								
1.	Overall understanding of the natural resources.							
2.	Basic understanding of the ecosystem and its diversity.							
3.	Acquaintance on various environmental challenges induced due to unplanned anthropogenic activities.							
4.	An understanding of the environmental impact of developmental activities.							
5.	Awareness on the social issues, environmental legislation and global treaties.							
SYLLABUS								
UNIT-I (8 Hrs)	Multidisciplinary nature of Environmental Studies: Definition, Scope and Importance – Sustainability: Stockholm and Rio Summit–Global Environmental Challenges: Global warming and climate change, acid rains, ozone layer depletion, population growth and explosion, effects;. Role of information technology in environment and human health. Ecosystems: Concept of an ecosystem. - Structure and function of an ecosystem; Producers, consumers and decomposers. - Energy flow in the ecosystem - Ecological succession. - Food chains, food webs and ecological pyramids; Introduction, types, characteristic features, structure and function of Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystems.							
UNIT-II (8 Hrs)	Natural Resources: Natural resources and associated problems. Forest resources: Use and over – exploitation, deforestation – Timber extraction – Mining, dams and other effects on forest and tribal people. Water resources: Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems. Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources. Food resources: World food problems, changes caused by non-agriculture activities-effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity. Energy resources: Growing energy needs, renewable and non-renewable energy sources use of alternate energy sources. Land resources: Land as a resource, land degradation, Wasteland reclamation, man induced landslides, soil erosion and desertification; Role of an individual in conservation of natural resources; Equitable use of resources for sustainable lifestyles.							
UNIT-III (8 Hrs)	Biodiversity and its conservation: Definition: genetic, species and ecosystem diversity-classification - Value of biodiversity: consumptive use, productive use, social-Biodiversity at national and local levels. India as a mega-diversity nation - Hot-spots of biodiversity - Threats to biodiversity: habitat loss, man-wildlife conflicts. - Endangered and endemic species of India – Conservation of biodiversity: conservation of biodiversity.							

UNIT-IV (8 Hrs)	Environmental Pollution: Definition, Cause, effects and control measures of Air pollution, Water pollution, Soil pollution, Noise pollution, Nuclear hazards. Role of an individual in prevention of pollution. - Pollution case studies, Sustainable Life Studies. Impact of Fire Crackers on Men and his well being. Solid Waste Management: Sources, Classification, effects and control measures of urban and industrial solid wastes. Consumerism and waste products, Biomedical, Hazardous and e – waste management.
UNIT-V (8 Hrs)	Social Issues and the Environment: Urban problems related to energy -Water conservation, rain water harvesting-Resettlement and rehabilitation of people; its problems and concerns. Environmental ethics: Issues and possible solutions. Environmental Protection Act -Air (Prevention and Control of Pollution) Act. –Water (Prevention and control of Pollution) Act -Wildlife Protection Act -Forest Conservation Act-Issues involved in enforcement of environmental legislation.-Public awareness.
UNIT-VI (8 Hrs)	Environmental Management: Impact Assessment and its significance various stages of EIA, preparation of EMP and EIS, Environmental audit. Ecotourism, Green Campus – Green business and Greenpolitics. The student should Visit an Industry / Ecosystem and submit a report individually on any issues related to Environmental Studies course and make a power point presentation.
Text Books:	
1.	Environmental Studies, K. V. S. G. Murali Krishna, VGS Publishers, Vijayawada Rani; Pearson Education, Chennai
2.	Environmental Studies, R. Rajagopalan, 2 nd Edition, 2011, Oxford University Press.
3.	Environmental Studies, P. N. Palanisamy, P. Manikandan, A. Geetha, and K. Manjula
Reference Books:	
1.	Text Book of Environmental Studies, Deeshita Dave & P. Udaya Bhaskar, Cengage Learning.
2.	A Textbook of Environmental Studies, Shaashi Chawla, TMH, New Delhi
3.	Environmental Studies, Benny Joseph, Tata McGraw Hill Co, New Delhi
4.	Perspectives in Environment Studies, Anubha Kaushik, C P Kaushik, New Age International Publishers, 2014

Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC2103	MC	2	--	--	--	--	--	--
ENGLISH PROFICIENCY								
(Common to CSE, ECE & IT)								
Course Objectives: The students will be able to								
1.	Communicate their ideas and views effectively							
2.	Practice language skills and improve their language competency.							
3.	Know and perform well in real life contexts							
4.	Identify and examine their self-attributes which require improvement and motivation.							
5.	Build confidence and overcome their inhibitions, stage freight, nervousness etc.,							
6.	Improve their reading skills.							
Course Outcomes: The students will								
S.No	Outcome							Knowledge Level
1.	Improve speaking skills.							K3
2.	Enhance their listening capabilities							K3
3.	Learn and practice the skills of composition writing.							K3
4.	Enhance their reading and understanding of different texts.							K3
5.	Improve their communication both in formal and informal contexts.							K3
6.	Be confident in presentation skills.							K3
SYLLABUS								
UNIT-I	Listening Skills Types of listening Hearing and Listening Listening as a receptive skill							
UNIT-II	Speaking Skills Presentation skills Describing event/place/thing Extempore Debate Group Discussion							
UNIT-III	Reading Skills Types of Reading (Intensive and Extensive reading, Skimming, Scanning) Reading/Summarizing News Paper Articles							
UNIT-IV	Writing Skills Essay Writing (Argumentative, Analytical and Descriptive) E-Mail Writing Business Letters Resume Writing							

UNIT-V	Integrated Language Skills Listening Skills for Speaking and Writing Reading Skills for Writing and Speaking
Reference Books:	
1.	Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
2.	English and Communication Skills for Students of Science and Engineering, by S.P. Dhanavel, Orient Blackswan Ltd. 2009
3.	Enriching Speaking and Writing Skills, Orient Blackswan Publishers.
4.	The Oxford Guide to Writing and Speaking by John Seely OUP.
5.	Effective Technical Communication by M.AshrafRizwi. Tata Mcgraw hill.
6.	Six Weeks to Words of Power by Wilfred Funk. W.R.Goyal Publishers
Note: Internal Assessment is carried out throughout the semester.	

Regulation: R20			II / IV - B.Tech. II - Semester						
COMPUTER SCIENCE & ENGINEERING									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards)									
Course Code	Course Name	Category	Cr	L	T	P	Int. Marks	Ext. Marks	Total Marks
B20 BS 2201	Probability and Statistics	BS	3	3	0	0	30	70	100
B20 CS 2201	Data Base Management Systems	PC	3	3	0	0	30	70	100
B20 CS 2202	Design and Analysis of Algorithms	PC	3	3	0	0	30	70	100
B20 CS 2203	Java Programming	ES	3	3	0	0	30	70	100
B20 HS 2201	Managerial Economics and Financial Accountancy	HS	3	3	0	0	30	70	100
B20 CS 2204	Data Base Management Systems Lab	PC	1.5	0	0	3	15	35	50
B20 CS 2205	R Programming Lab	PC	1.5	0	0	3	15	35	50
B20 CS 2206	Java Programming Lab	ES	1.5	0	0	3	15	35	50
B20 CS 2207	Applications of Python using Numpy and Pandas	SOC	2	0	0	4	--	50	50
TOTAL			21.5	15	0	13	195	505	700

Code	Category	L	T	P	C	I.M	E.M	Exam
B20 BS 2201	BS	3	0	0	3	30	70	3 Hrs.
PROBABILITY AND STATISTICS								
(Common to AIDS & CSE)								
Course Objectives:								
1.	Have an idea of data science and single and joint random variables.							
2.	Learn the concept of mathematical expectation, generating functions and their properties.							
3.	Fit a linear or nonlinear curve for a data using method of least squares.							
4.	Know about the correlation coefficient and regression lines.							
5.	Analyse various statistical measures of a few discrete and continuous probability distributions.							
6.	Develop a framework for testing of hypothesis in giving inferences about Population parameters.							
Course Outcomes:								
At the end of the course Students will be able								
1.	Understand the concepts of data science and identify a random variable as discrete/continuous and analyse it.							K2
2.	Determine statistical measures like Mean, Variance and generating functions in terms of Expectations.							K3
3.	Determine a best suitable curve for a given data using the method of least squares.							K3
4.	Determine correlation coefficient and regression lines.							K3
5.	Solve simple problems based on discrete and continuous probability distributions.							K3
6.	Apply testing of hypothesis for getting inferences about Population Parameters based on Sample statistic.							K3
SYLLABUS								
UNIT-I (12 Hrs)	Descriptive statistics and methods for data science: Data science, Statistics Introduction, Collection of data, primary and secondary data, Type of variables: dependent and independent, Categorical and Continuous variables, Data visualization, Measures of Central tendency, Measures of Variability (spread or variance), Moments, Measures of Skewness and Kurtosis. Random Variables and Probability functions: Definition of a random variable, Distribution function, Properties of Distribution Function, Discrete Random Variable, Probability Mass Function, Discrete Distribution Function, Continuous Random Variable, Probability Density Function, Continuous Distribution Function.							
UNIT-II (10 Hrs)	Two-dimensional random variables: Joint probability mass function and density functions, two-dimensional distribution function, marginal functions, simple examples. Mathematical Expectation: Mathematical Expectation of a Random Variable, Expected Value of function of a Random Variable, Addition Theorem and Multiplication Theorem of Expectation (without proofs), Statistical Measures like Mean, Variance, Moments and Covariance in terms of Expectations. Generating functions: Moment generating Function, Characteristic Function of a Random Variable and cumulant generating function.							

UNIT-III (12 Hrs)	Curve fitting: Method of least Squares, fitting of a Straight line, Fitting of a Parabola, fitting of Exponential curves: $y = ae^{bx}$, $y = ab^x$ and Power curve: $y = ax^b$ Correlation: Definition, Karl Pearson's Coefficient of Correlation, Limits for correlation coefficient, Rank Correlation, Spearman's formula for rank correlation coefficient (without proofs). Regression Analysis: Regression Lines, Regression Coefficients and their properties (without proofs).
UNIT-IV (12 Hrs)	Discrete and Continuous Distributions: Discrete Distributions: Uniform distribution, Binomial distribution and Poisson distribution - Mean, Variance, moments, m.g.f., Characteristic function, Fitting of distributions. Continuous Distributions: Uniform distribution, Normal Distribution - Standard Normal Variate, Mean, Variance, m.g.f., Characteristic function, cumulant generating function.
UNIT-V (12 Hrs)	Sampling theory and Testing of Hypothesis: Sampling Theory: Sample, population, statistic, parameter, Sampling distribution of a statistic, standard error, point and interval estimation. Testing of Hypothesis- Formulation of Null hypothesis, Alternative hypothesis, Critical region, level of significance, Errors in sampling- Type-I-error, Type-II-error, One-tailed and Two-tailed tests. Degrees of freedom. Large Sample Theory: Test of significance for single proportion and difference of proportions. Small Sample Theory: Student's-t-distribution: definition, t-test for single mean, t-test for difference of means, Paired t-test for difference of means. F-distribution: definition, F-test for equality of two population variances. Chi-square distribution: definition, Chi-square test for goodness of fit.
Text Books:	
1.	Probability, Statistics and Random Processes by T.Veerarajan, Tata Mc Graw Hill Pub.
2.	Fundamentals of Mathematical Statistics by S. C. Gupta and V.K. Kapoor, Sultan Chand & Sons Publishers.
Reference Books:	
1.	Higher Engineering Mathematics, by Dr.B.S.Grewal, 43 rd Edition, Khanna Publishers.
2.	Probability and statistics for Engineers, Miller and Freund, 7 th edition, Prentice-Hall India.
3.	Probability and statistics for Engineers and Scientists by Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye, Eighth edition, Pearson Education.
4.	Michael Baron, Probability and statistics for computer scientists(1 st edn.), Chapman and Hall Book, 2003.
	Paul L. Meyer, Introductory Probability and Statistical Applications (2 nd edn.), Addison-Wesley, 1970.
e-Resources:	
1.	http://www.swayam.gov.in

Code	Category	L	T	P	C	I.M	E.M	Exam
B20 CS 2201	PC	3	0	0	3	30	70	3 Hrs.
DATABASE MANAGEMENT SYSTEMS								
(For CSE)								
Course Objectives:								
1.	To introduce about database management systems							
2.	To give a good formal foundation on the relational model							
3.	To introduce the concepts of basic SQL as a universal Database language							
4.	To demonstrate the principles behind systematic database design approaches by covering conceptual design, logical design and normalization							
5.	To provide an overview of physical design of a database system, by discussing Database indexing techniques and storage techniques							
6.	To explain Transaction management techniques							
Course Outcomes:								
At the end of the course Students will be able to								
1.	Describe fundamental concepts a relational database							K2
2.	Create, maintain and manipulate a relational database using SQL							K3
3.	Apply Conceptual and Logical database design							K3
4.	Apply normalization for database design							K3
5.	Illustrate Storage management and Transaction management techniques.							K2
SYLLABUS								
UNIT-I (8Hrs)	Introduction: Database system, Characteristics (Database Vs File System), Database Users (Actors on Scene, Workers behind the scene), Advantages of Database systems, Database applications. Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture for data independence; Database system structure, environment, Centralized and Client Server architecture for the database.							
UNIT-II (10 Hrs)	Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update), basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions (Date and Time, Numeric, String conversion).							
UNIT-III (10 Hrs)	Entity Relationship Model: Introduction, Basic features of ER model, Representation of entities, attributes, entity set, relationship, relationship set, constraints, ER diagrams Generalization/specialization and Aggregation. SQL: Creating tables with relationships, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, ordering, implementation of different types of joins, views(updatable and non-updatable), relational set operations.							

UNIT-IV (10 Hrs)	Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, Closure of functional dependency and attribute closure, Normal forms based on functional dependency(1NF, 2NF and 3 NF), concept of surrogate key, Boyce-Codd normal form(BCNF), Lossless join and dependency preserving decomposition, Fourth normal form(4NF), Fifth Normal Form (5NF).
UNIT-V (10 Hrs)	Transaction Concept: Transaction State, Implementation of Atomicity and Durability, Schedules, Serializability, Recoverability, Implementation of Isolation levels, 2Pl and Time stamp ordering protocols, Failure Classification, Recovery and Atomicity, ARIES Recovery algorithm. Indexing Techniques: Indexing, Cluster Indexes, Primary and Secondary Indexes , Index data Structures, Hash Based Indexing, B+ Trees: Searching, Insertion, Deletion
Text Books:	
1.	Database System Concepts by Abraham Silberschatz, Henry F. Korth, S. Sudarshan, 7th Edition, McGraw-Hill Education, 2019.
2.	Database Management Systems by Raghu Ramakrishnan, Johannes Gehrke, 3rd Edition., McGraw-Hill Education (India), 2014.
Reference Books:	
1.	Database Principles: Fundamentals of Design, Implementation, and Management by Steven Morris, Keeley Crockett, Carlos Coronel, Craig Blewett, Cengage, 2020.
2.	Fundamentals of Database Systems by RamezElmasri, Shamkant B. Navathe, 7th Edition, Pearson Education India, 2015.
3.	Introduction to Database Systems by C J Date, 8th Edition, Pearson Education, 2009.
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105175/
2.	https://www.geeksforgeeks.org/introduction-to-nosql/

Code	Category	L	T	P	C	I.M	E.M	Exam
B20 CS 2202	PC	3	0	0	3	30	70	100
DESIGN AND ANALYSIS OF ALGORITHMS								
(For CSE)								
Course Objectives:								
1.	Analyze the asymptotic performance of algorithms							
2.	Write rigorous correctness proofs for algorithms.							
3.	Demonstrate a familiarity with major algorithms design paradigms and data structures							
4.	Synthesize efficient algorithms in common engineering design situations							
Course Outcomes:								
At the end of the course Students will be able								
1.	Apply mathematical analysis methods to analyse the algorithm running times using asymptotic notations							K3
2.	Compare and understand how the choice of data structures impact the performance of various greedy algorithms							K2
3.	Describe, apply and analyze the complexities of Dynamic Programming Algorithms							K3
4.	Describe, apply and analyze the complexity of Backtracking and Branch and Bound, and explain the situations which call for usage of these paradigms							K3
5.	Infer lower bounds for common problems like searching, sorting, merging, selection, Understand the concepts of P, NP classes							K2
SYLLABUS								
UNIT-I (14Hrs)	INTRODUCTION: Getting Started: Insertion sort, Analyzing algorithms, Designing algorithms, Growth of Functions: Asymptotic Notation, Standard notations and common functions, Master method for solving recurrences DIVIDE AND CONQUER: General method, Finding maximum and minimum, Merge sort, Quick sort, Performance Measurement, Selection Problem, A Worst-Case Optimal Algorithm, Strassen's matrix multiplication, Convex hull Problem- QuickHull Algorithm							
UNIT-II (10 Hrs)	Sets and Disjoint set union, Union and Find Operations THE GREEDY METHOD: General method, Knapsack problem, Job sequencing with deadlines, Minimum cost spanning trees- Prim's algorithm, Kruskal's algorithm, Optimal storage on tapes, Optimal merge patterns , Huffman coding, Single source shortest paths.							
UNIT-III (12Hrs)	DYNAMIC PROGRAMMING: General method, Multistage graphs, All pairs shortest paths, Optimal binary search trees, String editing, 0/1Knapsack, The travelling salesperson problem							

UNIT-IV (12Hrs)	BACKTRACKING: General method, 8-Queens problem, Sum of subsets, Graph colouring, Hamiltonian cycles BRANCH AND BOUND: The method, Least Cost (LC) Search, The 15-puzzle problem, Control abstractions for LC Search, FIFO Branch-and-Bound, LC Branch-and-Bound, 0/1 Knapsack problem: LC Branch and Bound, FIFO Branch-and-Bound, Travelling salesperson problem: LC Branch and Bound
UNIT-V (8Hrs)	LOWER BOUND THEORY: The method, Comparison trees, Ordered Searching, Sorting, Selection, Oracles and adversary arguments, Merging, Largest and Second largest NP-HARD AND NP-COMPLETE PROBLEMS: Basic concepts, Nondeterministic Algorithms, The Classes NP-hard and NP-complete, Cook's theorem. Modular Arithmetic.
Text Books:	
1.	Fundamentals of Computer Algorithms 2nd edition by Ellis Horowitz, Sartaj Sahni, S. Rajasekharan, university press, 2008
2.	Introduction to Algorithms 3rd edition by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, PHI, 2010
Reference Books:	
1.	Design and Analysis Algorithms - Parag Himanshu Dave, Himanshu Bhalchandra Dave Publisher: Dhanpat Rai & co, 2017
2.	Algorithm Design: Foundations, Analysis and Internet examples, M.T. Goodrich and R. Tomassia, John Wiley and sons, 2006
3.	Introduction to the Design and Analysis of Algorithms, Anany Levitin, PEA, 3rd Edition.
4.	Foundations of Algorithms, R. Neapolitan and K. Naimipour, Jones and Bartlett, Pearson, 2011
5.	Advanced Data Structures – Peter Brass, Cambridge University Press, 2008
e-Resources:	
1.	https://onlinecourses.nptel.ac.in/noc19_cs47/preview
2.	https://onlinecourses.swayam2.ac.in/cec20_cs03/preview
3.	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-046j-design-and-analysis-of-algorithms-spring-2015/

Code	Category	L	T	P	C	I.M	E.M	Exam
B20 CS 2203	ES	3	0	0	3	30	70	3 Hrs.
JAVA PROGRAMMING								
(For CSE)								
Course Objectives:								
1.	To identify Java language components and how they work together in applications							
2.	To learn the fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries.							
3.	To learn how to extend Java classes with inheritance and dynamic binding and how to use exception handling in Java applications							
4.	To understand how to design applications with threads in Java.							
5.	To understand how to use Java JDBC APIs for program development.							
Course Outcomes:								
At the end of the course Students will be able								
1.	Recall the syntax and semantics of java programming language and basic concepts of OOP.							K1
2.	Relate array data structure and string manipulation operations							K2
3.	Develop reusable programs using the concepts of inheritance, interfaces and packages							K3
4.	Apply the concept of Exception handing and multithreading to build an efficient and error free code.							K3
5.	Develop a program that manages input & output streams and apply JDBC to interface with database.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to JAVA: Structure of JAVA, Features of JAVA, JAVA Tokens, Command Line Arguments, Operators, Type Casting, Control Statements. Classes & Objects, Constructors: Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, method overloading. Constructors: Default Constructor, Parameterized Constructor, Copy Constructor and Constructor Overloading, This Keyword.							
UNIT-II (10 Hrs)	Arrays& Vectors: Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Operations on Array Elements, Assigning Array to Another Array, Two-dimensional Arrays, Three-dimensional Arrays, Vectors: Introduction to vectors, Vector methods. String Handling in Java: Introduction, methods in String Class, Methods for comparison of Strings, StringBuffer class.							
UNIT-III (10Hrs)	Inheritance: Introduction, Types of Inheritances, Single inheritance, Multi-level inheritance, Hierarchical Inheritance, Hybrid Inheritance and Multiple inheritance,Method Overriding, Super Keyword, Final Keyword, Abstract Classes. Interfaces and Packages: Introduction, Declaration of Interface, Implementing Interfaces, Extending interfaces, Default Methods in Interfaces, Static Methods in Interface. Packages: Introduction, Defining Package, Importing Packages, Access Modifiers.							

UNIT-IV (10Hrs)	Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Statements, Class Throwable, Custom Exceptions, Nested try and catch Blocks. Multithreaded Programming: Introduction, Need for Multiple Threads, Thread class, Thread Life Cycle, Extending Thread class, Implementing Runnable interface, Thread Priorities, Inter-thread communication.
UNIT-V (10Hrs)	File IO: Introduction, IO classes and interfaces, Stream classes, Byte Streams, the character Streams. Java Database Connectivity: Introduction, Structure of JDBC, JDBC Architecture, Types of JDBC Drivers, JDBC API (java.sql package), Connecting to the Database, JAVA Database connection program for MS-Access, Oracle and MySQL, JDBC Batch Processing.
Text Books:	
1.	Core Java Volume I--Fundamentals: 1 (Core Series)11 th edition (2020) by Cay Horstmann, Publisher: Pearson
2.	The complete Reference Java, 12th edition (2021), Herbert Schildt, Publisher:TMH.
3.	JdbcApiTutorial and Reference 3E(2003), by Maydene, Jon Ellis (Author), Jonathan Bruce, Publisher : Addison-Wesley Professional
Reference Books:	
1.	Introduction to java programming, 9th edition(2014) by Y Daniel Liang, Publisher:Pearson
2.	Murach's Java Programming, 5 th edition(2017) Joel Murach , Publisher:Mike <i>Murach</i>
3.	JAVA one step ahead, 1 st edition (2017)Anitha Seth, B.L.Juneja, Oxford.
4.	Java: A Beginner's Guide, Eighth Edition 8th Edition(2018)by Herbert Schildt,Publisher:McGraw-Hill Education
5.	Head First Java 3e (2021)(A Brain Friendly Guide) by Kathy Sierra & Bert bates,Publisher:O'Reilly
6.	Programming With Java:A Primer 6E(2019)By Balagurusamy,Publisher:TMH.
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105191/
2.	https://www.coursera.org/learn/java-introduction
3.	https://docs.oracle.com/javase/tutorial/
4.	https://www.linkedin.com/in/jamesgosling
5.	https://en.wikipedia.org/wiki/James_Gosling#Books

Code	Category	L	T	P	C	I.M	E.M	Exam
B20 HS 2201	HS	3	0	0	0	30	70	3 Hrs.
MANAGERIAL ECONOMICS AND FINANCIAL ACCOUNTANCY								
(For CSE)								
Course Objectives:								
1.	To Study Managerial Economics and Demand Analysis							
2.	To familiarize about the Concepts of Cost and Break-Even Analysis.							
3.	To understand the nature of markets and to know the Pricing Policies							
4.	To learn about accounting cycle and preparation of Financial Statements.							
5.	To know the concept of Capital and sources of raising and Depreciation							
Course Outcomes:								
At the end of the course Students will be able								
1.	Equip oneself with the knowledge of estimating the Demand and demand elasticities for a product.							K2
2.	Have knowledge of Cost and its types and ability to calculate BEP							K3
3.	Understand the nature of different markets							K2
4.	Uunderstand Pricing Practices prevailing in today's business world							K2
5.	Prepare Financial Statements and know how to calculate Profit & Loss for a firm							K3
6.	Know Types of capital and their sources and know how to calculate Depreciation							K2
SYLLABUS								
UNIT-I (10 Hrs)								
Introduction to Managerial Economics and demand Analysis: Managerial Economics: Definition of Economics & Classification of Economics (Micro & Macro), Meaning, Nature, & Scope of Managerial Economics.Demand Analysis: Concept of Demand, Determinants of Demand, Demand schedule, Demand curve, Law of Demand and its exceptions. Elasticity of Demand, Types of Elasticity of Demand. Importance of demand forecasting and its Methods.								
UNIT-II (10 Hrs)								
Cost Analysis: Importance of cost analysis, Types of Cost- Actual cost Vs Opportunity cost, Fixed cost Vs Variable cost, Explicit Vs Implicit cost, Historical cost Vs Replacement cost, Incremental cost Vs Sunk cost; Elements of costs – Material, Labour, Expenses; Methods of costing - Job costing, contract costing, Process costing, Batch costing, Unit costing, Service costing, Multiple costing. Break-even analysis: Determination of Breakeven point - Applications, Assumptions and Limitations of Break -even analysis (Theory only).								
UNIT-III (10 Hrs)								
Introduction to Markets & Pricing Policies Market Structures: Salient Features of Perfect Competition, Monopoly, Monopolistic competition, Oligopoly and Duopoly. Pricing: Importance of pricing and its meaning;Methods of Pricing: Cost Based -Full cost, Mark-up, Marginal &Break-even; Demand Based - Penetrating, Skimming; Competition Based- Going rate, Sealed Bid, Discount; Internet Pricing - Flat-rate, Usage sensitive.								

UNIT-IV (08 Hrs)	Introduction to Financial Accounting: Importance of Accounting - Double Entry System of Accounting - Types of Accounts - Journal, Ledger, Trail Balance, Trading Account, Profit and Loss Account and Balance Sheet (outlines only).
UNIT-V (12 Hrs)	Capital & Depreciation: Types of Capital - Fixed capital & Working Capital, Components of Working Capital, Factors influencing Working capital. Methods of Raising Finance - Short term, medium term and Long term. Depreciation - Meaning, Importance and causes of depreciation; Methods of Depreciation- Straight line and diminishing balancing methods (Theory only)
Text Books:	
1.	A R Aryasri, Managerial Economics and Financial Analysis, TMH Pvt. Ltd, New Delhi
2.	Dr. N.Appa Rao, Dr.P. Vijayakumar:Managerial Economics and Financial Analysis', Cengage Publications, New Delhi
Reference Books:	
1.	Dr.B.Kuberudu& T.V. Ramana : Managerial Economics and Financial anaysis, Himalaya Publishing House
2.	Varshney R.L, K.L Maheswari, Managerial Economics, S. Chand & Company Ltd,
3.	Shashi K. Gupta & R.K. Sharma Management Accounting, Kalyani Publishers
4.	Maheswari S.N, An Introduction to Accountancy, Vikas Publishing House Pvt Ltd

Code	Category	L	T	P	C	I.M	E.M	Exam
B20 CS 2204	PC	0	0	3	1.5	15	35	3 Hrs.
DATABASE MANAGEMENT SYSTEMS LAB								
(For CSE)								
Course Objectives:								
1.	Populate and query a database using SQL DDL/DML Commands							
3.	Declare and enforce integrity constraints on a database							
4.	Writing Queries using advanced concepts of SQL							
5.	Programming PL/SQL including procedures, functions, cursors and triggers							
Course Outcomes:								
At the end of the course Students will be able to								
1.	Utilize SQL to execute queries for creating database and performing data manipulation operations							K3
2.	Apply Queries using SQL							K3
4.	Build PL/SQL programs including stored procedures, functions, cursors and triggers							K3
LIST OF PROGRAMS								
1	Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.							
2	Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET, Constraints. Example:- Select the roll number and name of the student who secured fourth rank in the class.							
3	Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.							
4	Queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)							
5	i. Create a simple PL/SQL program which includes declaration section, executable section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found) ii. Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.							
6	Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.							
7	Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, User defined Exceptions, RAISE APPLICATION ERROR.							
8	Programs development using creation of procedures, passing parameters IN and OUT of PROCEDURES.							
9	Program development using creation of stored functions, invoke functions in SQL Statements and write complex functions.							
10	Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE							

	CURRENT of clause and CURSOR variables.
11	Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers
Reference Books:	
1.	Oracle Database 12C: The Complete Reference by Byrla, McGraw Hill Education, 2017.
2.	SQL The Complete Reference by James Groff, Paul Weinberg , Andy Oppel, 3rd Edition, McGraw Hill Education, 2017.
3.	SQL, PL/SQL by Ivan Bayross, 4th Revised Edition, 2020.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20 CS 2205	PC	0	0	3	1.5	15	35	3 Hrs.
R PROGRAMMING LAB								
(For CSE)								
Course Objectives:								
1.	To learn statistical programming, computation, graphics, and modeling,							
2.	To learn Writing functions and use R in an efficient way,							
3.	To learn about basic types of statistical models							
Course Outcomes:								
At the end of the course Students will be able								
1.	Access online resources for R and import new function packages into the Rworkspace							K2
2.	Import, review, manipulate and summarize data-sets in R							K3
3.	Explore data-sets to create testable hypotheses and identify appropriate statistical tests							K3
4.	Perform appropriate statistical tests using R							K3
5.	Create and edit visualizations with R							K3
LIST OF PROGRAMS								
1.	Write a R program to take input from the user (name and age) and display the values. Also print the version of R installation.							
2.	Write a R program to get the details of the objects in memory.							
3.	Write a R program to create a sequence of numbers from 20 to 50 and find the mean of numbers from 20 to 60 and sum of numbers from 51 to 91.							
4.	Write a R program to create a simple bar plot of five subjects marks.							
5.	Write a R program to get the unique elements of a given string and unique numbers of vector.							
6.	Write a R program to create three vectors a,b,c with 3 integers. Combine the three vectors to become a 3×3 matrix where each column represents a vector. Print the content of the matrix.							
7.	Write a R program to create a 5 x 4 matrix, 3 x 3 matrix with labels and fill the matrix by rows and 2 × 2 matrix with labels and fill the matrix by columns.							
8.	Write a R program to combine three arrays so that the first row of the first array is followed by the first row of the second array and then first row of the third array.							
9.	Write a R program to create a two-dimensional 5x3 array of sequence of even integers greater than 50.							
10.	Write a R program to create an array using four given columns, three given rows, and two given tables and display the content of the array.							
11.	Write a R program to create an empty data frame.							
12.	Write a R program to create a data frame from four given vectors.							
13.	Write a R program to create a data frame using two given vectors and display the duplicated elements and unique rows of the said data frame.							
14.	Write a R program to save the information of a data frame in a file and display the information of the file.							
15.	Write a R program to create a matrix from a list of given vectors.							
16.	Write a R program to concatenate two given matrices of same column but different rows.							

17.	Write a R program to find row and column index of maximum and minimum value in a given matrix.
18.	Write a R program to append value to a given empty vector.
19.	Write a R program to multiply two vectors of integers type and length 3.
20.	Write a R program to find Sum, Mean and Product of a Vector, ignore element like NA or NaN.
21.	Write a R program to list containing a vector, a matrix and a list and give names to the elements in the list.
22.	Write a R program to create a list containing a vector, a matrix and a list and give names to the elements in the list. Access the first and second element of the list.
23.	Write a R program to create a list containing a vector, a matrix and a list and remove the second element.
24.	Write a R program to select second element of a given nested list.
25.	Write a R program to merge two given lists into one list.
26.	Write a R program to create a list named s containing sequence of 15 capital letters, starting from 'E'.
27.	Write a R program to assign new names "a", "b" and "c" to the elements of a given list.
28.	Write a R program to find the levels of factor of a given vector.
29.	Write a R program to create an ordered factor from data consisting of the names of months.
30.	Write a R program to concatenate two given factor in a single factor.
Reference Books:	
1.	R for Data Science is a book written by Hadley Wickham (Author), Garrett Grolemund.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20 CS 2206	ES	0	0	3	1.5	15	35	3 Hrs.
JAVA PROGRAMMING LAB								
(For CSE)								
Course Objectives:								
1.	Practice programming in the Java							
2.	Gain knowledge of object-oriented paradigm in the Java programming language							
3.	Learn use of Java in a variety of technologies and on different platforms							
Course Outcomes:								
At the end of the course Students will be able								
1.	Develop simple programs using command line arguments, arrays, vectors and strings.							K3
2.	Demonstrate Classes, Objects, Constructors, Methods and Runtime Polymorphism.							K2
3.	Develop reusable programs using the concepts of inheritance, interfaces and packages.							K3
4.	Develop Applications using exception handing and multithreading.							K3
5.	Apply the concepts of Java IO Files and database in real time problem solving.							K3
LIST OF PROGRAMS								
1.	Exercise - 1 (Basics)							
	a) Write a JAVA program to display default value of all primitive data types of JAVA. b) Write a JAVA program using Command line arguments							
2.	Exercise - 2 (Classes, Objects, Constructors)							
	a) Write a JAVA program to implement Classes and Objects. b) Write a JAVA program implement method overloading. c) Write a JAVA program to implement constructor overloading. d) Write a JAVA program to implement this keyword.							
3.	Exercise - 3 (Arrays, Vectors, Strings)							
	a) Write a JAVA program to implement Matrix Multiplication. b) Write a JAVA program to implement Vector Operations. c) Write a JAVA program to implement String Operations. d) Write a JAVA program to implement StringBuffer class.							
4.	Exercise - 4(Inheritance)							
	a) Write a JAVA program to implement Single Inheritance. b) Write a JAVA program to implement multilevel Inheritance. c) Write a java program for abstract class to find areas of different shapes. d) Write a JAVA program to implement “super” keyword.							
5.	Exercise - 5 (Interfaces, Packages)							
	a) Write a JAVA program to implement Interface. b) Write a JAVA program to implement simple Packages. c) Write a JAVA program to access package from another package. d) Write a JAVA program to implement sub Packages.							

6.	Exercise - 6 (Exception Handling) - Write a JAVA program to implement the following Built in Exceptions. - Arithmetic Exception. - Array Index Out Of Bounds Exception - Number Format Exception. - Null Pointer Exception. - Write a JAVA program to implement multiple catch statements. - Write a JAVA program to implement user defined Exception.
7.	Exercise – 7 (Multithreading) - Write a JAVA program that creates threads by extending Thread class .First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds . - Write a JAVA program to implement Runnable Interface. - Write a program to implement priorities to Thread. - Write a JAVA program to implement Thread Synchronization(Multiplication tables)
8.	Exercise - 8 (File IO) - Write a JAVA program to copy contents of file into another using Byte Oriented IO. - Write a JAVA program to copy contents of file into another using Character Oriented IO. - Write a JAVA program to display contents of file using Line Oriented IO. - Write a JAVA program to convert the values into tokens using Scanner class.
9.	Exercise - 9 (JDBC) - Write a JDBC program to insert data into database. - Write a JDBC program to delete data from database. - Write a JDBC program to update data into database. - Write a JDBC program to retrieve data from database.
Reference Books:	
1.	Core Java Volume I--Fundamentals: 1 (Core Series) 11 th Edition(2020) by Cay Horstmann, Publisher: Pearson
2.	The complete Reference Java, 12th edition (2021), Herbert Schildt, Publisher:TMH.
3.	JdbcApi Tutorial and Reference 3 rd Edition(2003), by Maydene , Jon Ellis (Author), Jonathan Bruce, Publisher : Addison-Wesley Professional
4.	Introduction to java programming, 9th Edition(2014) by Y Daniel Liang, Publisher:Pearson
5.	Murach's Java Programming, 5 th Edition(2017) Joel Murach , Publisher:Mike <i>Murach</i>
6.	JAVA one step ahead, First Edition (2017)Anitha Seth, B.L.Juneja, Oxford.
7.	Programming With Java:A Primer 6 th Edition(2019) By Balagurusamy, Publisher:TMH.
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105191/
2.	https://www.coursera.org/learn/java-introduction
3.	https://docs.oracle.com/javase/tutorial/
4.	https://www.linkedin.com/in/jamesgosling
5.	https://en.wikipedia.org/wiki/James_Gosling#Books
6.	https://nptel.ac.in/courses/106/105/106105191/

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CS2207	SOC	--	--	4	2	--	50	3 Hrs.
APPLICATIONS OF PYTHON USING NUMPY AND PANDAS								
(For CSE)								
MODULE 1 : NUMPY								
Course Objectives: The objective of this lab is to								
1.	The objective of this lab is to acquire programming skills in Python package NumPy and perform mathematical and statistical operation							
Course Outcomes:								
At the end of the course Students will be able								
1.	Explain how data is collected, managed and stored for processing							K3
2.	Understand the workings of various numerical techniques, different descriptive measures of Statistics, correlation and regression to solve the engineering problems							K3
3.	Understand how to apply some linear algebra operations to n-dimensional arrays							K3
4.	Use NumPy perform common data wrangling and computational tasks in Python.							K3
LIST OF EXPERIMENTS								
1	NumPy Installation using different scientific python distributions(Anaconda, Python(x,y), WinPython, Pyzo)							
2	NumPy Basics (np.array, np.arange, np.linspace, np.zeros, np.ones, np.random.random, np.empty)							
3	Arrays (array.shape, len(array), array.ndim, array.dtype, array.astype(type), type(array))							
4	Array Manipulation (np.append, np.insert, np.resize, np.delete, np.concatenate, np.vstack, np.hstack)							
5	Mathematical Operations(np.add, np.subtract, np.divide, np.multiply, np.sqrt, np.sin, np.cos, np.log, np.dot, np.roots) , Statistical Operations(np.mean, np.median, np.std, array.corrcoef())							
6	NumPy data types							
7	NumPyndarray							
8	NumPy String Operations							
9	NumPy Financial functions							
10	NumPy Functional Programming							

MODULE 2 : PANDAS

Course Objectives:

- | | |
|----|--|
| 1. | To Understand The Fundamentals Of The Pandas Library In Python |
| 2. | How It Is Used To Handle Data And Also Develop Basic Skills In Data Analysis And Visualization |

Course Outcomes:

At the end of the course Students will be able

- | | | |
|----|---|----|
| 1. | Use Pandas to create and manipulate data structures like Series and DataFrames. | K3 |
| 2. | Work with arrays, queries, and dataframes | K3 |
| 3. | Query DataFrame structures for cleaning and processing and manipulating files | K3 |
| 4. | Understand best practices for creating basic charts | K3 |

LIST OF PROGRAMS

- | | |
|----|---|
| 1. | Perform the following:
1) Pandas Installation
2) Creating DataFrames |
| 2. | <p>A) Pandas DataSeries:</p> <p>1) Write a Pandas program to create and display a one-dimensional array-like object containing an array of data using Pandas module.</p> <p>2) Write a Pandas program to convert a Panda module Series to Python list and it's type.</p> <p>3) Write a Pandas program to add, subtract, multiple and divide two Pandas Series.</p> <p>4) Write a Pandas program to convert a NumPy array to a Pandas series.</p> <p>Sample Series:
NumPy array:
[10 20 30 40 50]
Converted Pandas series:
0 10
1 20
2 30
3 40
4 50
dtype: int64</p> |
| 3. | <p>B) Pandas DataFrames:</p> <p>Consider Sample Python dictionary data and list labels:
exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']</p> <p>1) Write a Pandas program to create and display a DataFrame from a specified dictionary data which has the index labels.</p> |

	2) Write a Pandas program to change the name 'James' to 'Suresh' in name column of the DataFrame. 3) Write a Pandas program to insert a new column in existing DataFrame. 4) Write a Pandas program to get list from DataFrame column headers. 5) Write a Pandas program to get list from DataFrame column headers.																																																								
4.	C) Pandas Index: 1) Write a Pandas program to display the default index and set a column as an Index in a given dataframe. 2) Write a Pandas program to create an index labels by using 64-bit integers, using floating-point numbers in a given dataframe.																																																								
5.	D) Pandas String and Regular Expressions: 1) Write a Pandas program to convert all the string values to upper, lower cases in a given pandas series. Also find the length of the string values. 2) Write a Pandas program to remove whitespaces, left sided whitespaces and right sided whitespaces of the string values of a given pandas series. 3) Write a Pandas program to count of occurrence of a specified substring in a DataFrame column. 4) Write a Pandas program to swap the cases of a specified character column in a given DataFrame.																																																								
6.	E) Pandas Joining and merging DataFrame: 1) Write a Pandas program to join the two given dataframes along rows and assign all data. 2) Write a Pandas program to append a list of dictionaries or series to a existing DataFrame and display the combined data. 3) Write a Pandas program to join the two dataframes with matching records from both sides where available.																																																								
7.	F) Pandas Time Series: 1) Write a Pandas program to create a) Datetimeobject for Jan 15 2012. b) Specific date and time of 9:20 pm. c) Local date and time. d) A date without time. e) Current date. f) Time from a datetime. g) Current local time. 2) Write a Pandas program to create a date from a given year, month, day and another date from a given string formats. 3) Write a Pandas program to create a time-series with two index labels and random values. Also print the type of the index.																																																								
8.	G) Pandas Grouping Aggregate: Consider dataset: <table><tr><th>school</th><th>class</th><th>name</th><th>date_Of_Birth</th><th>age</th><th>height</th><th>weight</th><th>address</th></tr><tr><td>S1</td><td>s001</td><td>V</td><td>Alberto Franco</td><td>15/05/2002</td><td>12</td><td>173</td><td>35 street1</td></tr><tr><td>S2</td><td>s002</td><td>V</td><td>Gino Mcneill</td><td>17/05/2002</td><td>12</td><td>192</td><td>32 street2</td></tr><tr><td>S3</td><td>s003</td><td>VI</td><td>Ryan Parkes</td><td>16/02/1999</td><td>13</td><td>186</td><td>33 street3</td></tr><tr><td>S4</td><td>s001</td><td>VI</td><td>Eesha Hinton</td><td>25/09/1998</td><td>13</td><td>167</td><td>30 street1</td></tr><tr><td>S5</td><td>s002</td><td>V</td><td>Gino Mcneill</td><td>11/05/2002</td><td>14</td><td>151</td><td>31 street2</td></tr><tr><td>S6</td><td>s004</td><td>VI</td><td>David Parkes</td><td>15/09/1997</td><td>12</td><td>159</td><td>32 street4</td></tr></table>	school	class	name	date_Of_Birth	age	height	weight	address	S1	s001	V	Alberto Franco	15/05/2002	12	173	35 street1	S2	s002	V	Gino Mcneill	17/05/2002	12	192	32 street2	S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33 street3	S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30 street1	S5	s002	V	Gino Mcneill	11/05/2002	14	151	31 street2	S6	s004	VI	David Parkes	15/09/1997	12	159	32 street4
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	1) Write a Pandas program to split the following dataframe into groups based on school code. Also check the type of GroupBy object. 2) Write a Pandas program to split the following dataframe by school code and get mean, min, and max value of age for each school.
9.	H) Pandas Styling: 1) Create a dataframe of ten rows, four columns with random values. Write a Pandas program to highlight the negative numbers red and positive numbers black. 2) Create a dataframe of ten rows, four columns with random values. Write a Pandas program to highlight the maximum value in each column. 3) Create a dataframe of ten rows, four columns with random values. Write a Pandas program to highlight dataframe's specific columns.
10.	I) Excel: 1) Write a Pandas program to import excel data into a Pandas dataframe. 2) Write a Pandas program to find the sum, mean, max, min value of a column of file.
11.	J) Plotting: 1) Write a Pandas program to create a horizontal stacked bar plot of opening, closing stock prices of any stock dataset between two specific dates. 2) Write a Pandas program to create a histograms plot of opening, closing, high, low stock prices of stock dataset between two specific dates. 3) Write a Pandas program to create a stacked histograms plot of opening, closing, high, low stock prices of stock dataset between two specific dates with more bins.
12.	K) Pandas SQL Query: 1) Write a Pandas program to display all the records of a student file. 2) Write a Pandas program to select distinct department id from employees file.
Reference Books:	
1	Wesley J Chen, Core Python Programming 2nd Edition, Kindle Edition
2	Jake VanderPlas, Python Data Science Handbook: Essential Tools for Working with Data 1st Edition, Kindle Edition, Oreilly Publisher
3	Mark Summerfield, Programming in Python 3-A Complete Introduction to the Python Language, Second Edition, Addison Wesley
4	Phuong Vo.T.H, Martin Czygan, Getting Started with Python Data Analysis, Packt Publishing Ltd
5	Ivan Idris, Python Data Analysis, Packt Publishing Ltd
6	Magnus Vilhelm Persson and Luiz Felipe Martins, Mastering Python Data Analysis, Packt Publishing Ltd